equator, carrying with you an arrow pointing ahead. Once you reach the equator, turn to the left without changing the direction of the arrow. The arrow still points south, which is now to your right. Advance a little towards the east along the equator and then turn again towards the north – again without changing the direction of the arrow, which will now be pointing behind you. When you reach the North Pole again, you have executed a closed circuit – a 'loop', as it is termed – and the arrow does not point in the same direction as when you started out (figure 3.14). The angle through which the arrow has turned in the course of the loop measures the curvature.

I will return later to this method of measuring curvature by making a loop in space. These will be the loops that give the name to the theory of loop quantum gravity.

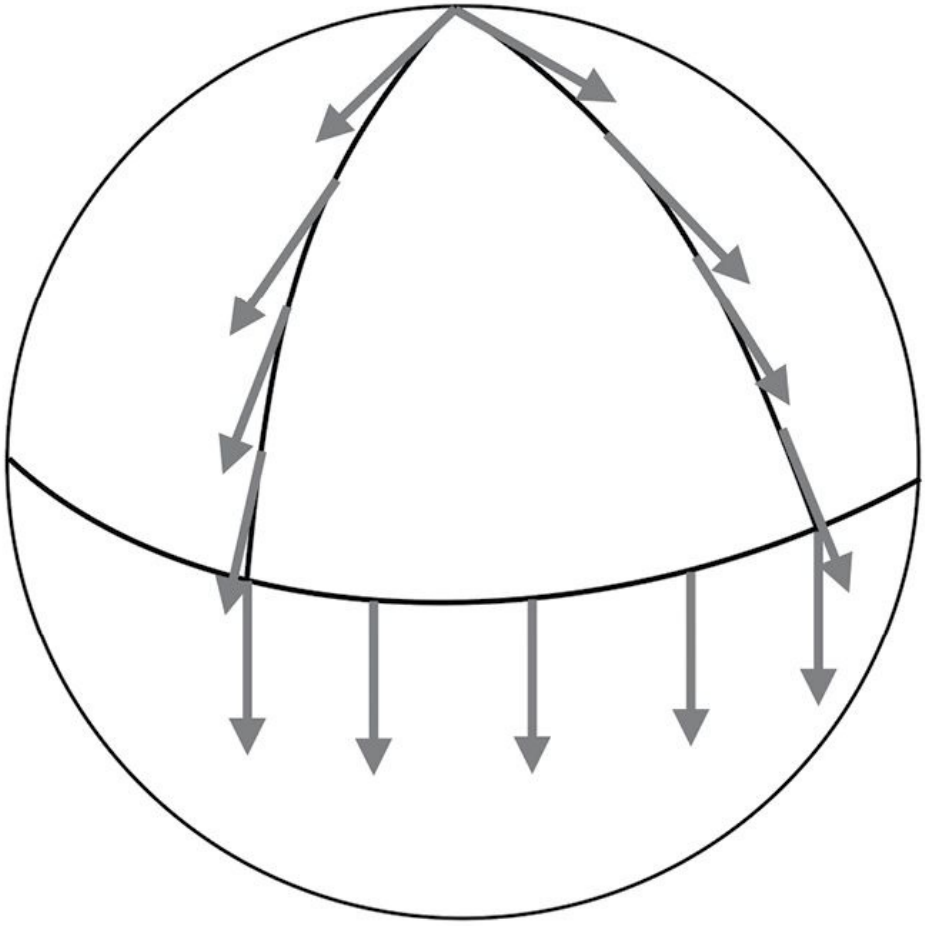


Figure 3.14 An arrow carried parallel to itself along a circuit (a loop) in a curved space arrives back rotated at the point of departure.

Dante leaves Florence in 1301, while the mosaics in the cupola of the Baptistery are being completed. The mosaic, representing Hell (the work of